MTH2003: Examples Sheet 1 Solutions

1. (a) **Question** $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 2 - 3y$.

Solution

[i] This equation is separable. To solve, write, for $y \neq 2/3$: $\dfrac{\mathrm{d}y}{2-3y} = \mathrm{d}t$, so that $\displaystyle\int \frac{\mathrm{d}y}{2-3y} = \int \mathrm{d}t \Rightarrow -\frac{1}{3}\ln|2-3y| = t + A$ and $|2-3y| = \mathrm{e}^{-3A}\,\mathrm{e}^{-3t}$. Thus $2 - 3y = B\,\mathrm{e}^{-3t}$ and $y = \left(2 - B\,\mathrm{e}^{-3t}\right)/3$, $t \in \mathbb{R}$, where $B = \pm\mathrm{e}^{-3A}$. Note that the special solution $y = 2/3$ can be obtained by the same formula by taking $B = 0$.

[ii] This equation is not homogeneous, as can be seen from $f(t, y) = 2 - 3y$ with $t = 1, y = 1, w = 2$ giving

$$f(wt, wy) = -4, \quad f(t, y) = -1$$

for example.

[iii] This equation is linear, $\mathrm{d}y/\mathrm{d}t + 3y = 2$. Here $p(t) = 3$ and $\int p(t)\mathrm{d}t = 3t + c$, so the integrating factor is $h(t) = \mathrm{e}^{3t}$. Then, multiplying the ODE by h gives $\dfrac{\mathrm{d}}{\mathrm{d}t}\left(\mathrm{e}^{3t}y\right) = \mathrm{e}^{3t}\dfrac{\mathrm{d}y}{\mathrm{d}t} + 3\,\mathrm{e}^{3t}y = 2\,\mathrm{e}^{3t}$ and integrating gives $y\,\mathrm{e}^{3t} = \displaystyle\int 2\,\mathrm{e}^{3t} = \frac{2}{3}\,\mathrm{e}^{3t} + A \Rightarrow y = \frac{2}{3} + A\,\mathrm{e}^{-3t}$, $t \in \mathbb{R}$, in agreement with [i].

(b) **Question** $\dfrac{1}{y}\dfrac{\mathrm{d}y}{\mathrm{d}t} - \dfrac{1}{t} = 0$.

Solution

[i] The standard form is $\dfrac{\mathrm{d}y}{\mathrm{d}t} = \dfrac{y}{t}$ which is separable. For $y \neq 0$ we have $\mathrm{d}y/y = \mathrm{d}t/t \Rightarrow \ln(|y|) = \ln(|t|) + C_1 \Rightarrow e^{\ln(|y|)} = e^{\ln(|t|)}e^{C_1} \Rightarrow |y| = e^{C_1}|t|$. Hence $y = \pm e^{C_1}t = C_2 t$, where $t \in (-\infty, 0)$ or $t \in (0, \infty)$. By construction $C_2 \neq 0$ in the typical solution. The standard form of the equation also has the special solution $y = 0$, which is covered by the typical solution formula if we allow $C_2 = 0$. However, $y = 0$ is disallowed for the original ODE because y is in the denominator.

[ii] The equation $\dfrac{\mathrm{d}y}{\mathrm{d}t} = \dfrac{y}{t}$ is homogeneous as the right-hand depends only on the combination y/t (in fact it just *is* that combination). So we do the standard substitution $y = tv$, which gives $dy/\mathrm{d}t = t\mathrm{d}v/\mathrm{d}t + v$. Substituting into the ODE leads to $t\mathrm{d}v/\mathrm{d}t = 0$. Hence $v = C$ and $y = Ct$ as before.

[iii] We can write $\mathrm{d}y/\mathrm{d}t - y/t = 0$, and the equation is linear with $p(t) = -1/t$. Hence the integrating factor is $h(t) = \exp(\int p(t)\,\mathrm{d}t) = \exp(-\ln(t)) = t^{-1}$. Multiplying the equation by that factor we get $\dfrac{1}{t}y' - \dfrac{y}{t^2} = 0$. We notice that the left-hand side is the full derivative $(y/t)' = 0$, and hence $y/t = C$ and $y = Ct$ as before.

Note that no matter which method is used, in the final answer we must exclude $t = 0$ from the intervals where the solutions are defined, and $y = 0$ from the set of the solutions, for the simple reason that both t and y appear in the denominators in the given ODE.

(c) **Question** $\dfrac{\mathrm{d}y}{\mathrm{d}t} = t^3 - y^3$.

Solution

[i] This equation is not separable because $t^3 - y^3$ can not be factorised into a product of a function of t and a function of y. Or, since we have $f(t, y) = t^3 - y^3$, taking $a = 0, b = 1, c = 1, d = 2$, for example, gives

$$f(a, b)f(c, d) = 7, \quad f(a, d)f(c, b) = 0$$

Hence $f(a,b)f(c,d) \neq f(a,d)f(c,b)$.

[ii] This equation is not homogeneous, as can be seen by taking $t = 2$, $y = 1$ and $w = 2$, giving

$$f(wt, wy) = f(4,2) = 4^3 - 2^3 = 56, \quad f(t,y) = f(2,1) = 7\,.$$

Hence $f(wt, wy) \neq f(t,y)$.

[iii] This equation is not linear due to the y^3 term.

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2. (a) **Question** $y' = ty$, *(A)* $y(1) = 1$, *(B)* $y(1) = 0$, *(C)* $y(0) = 1$, *(D)* $y(0) = 0$.

Solution

[i] The right-hand side $f(t,y) = ty$ and its derivative $f_y(t,y) = t$ are continuous for all t and all y. Therefore, the IVP will be locally solvable and locally uniquely solvable for any initial condition, $\boxed{(t_0, y_0) \in \mathbb{R} \times \mathbb{R}}$.

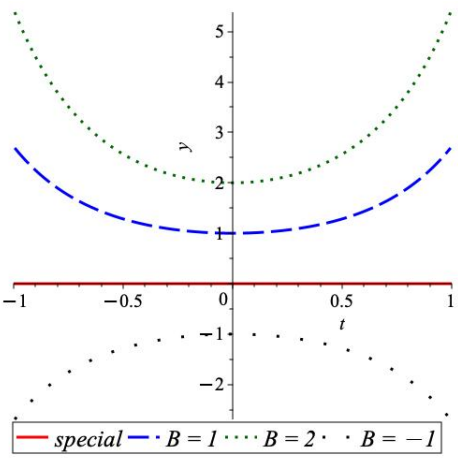
[ii] The right-hand side factorises $f(t,y) = ty = F(t)G(y)$ with $F(t) = t$ and $G(y) = y$. We have $G(y) = 0$ for $y = 0$ and $\boxed{y(t) \equiv 0}$ is a special solution. It is defined for all t (as it doesn't even depend on t). Otherwise, for $y \neq 0$, we get the typical solutions by separating the variables, $y^{-1}\mathrm{d}y = t\,\mathrm{d}t \Rightarrow \ln|y| = t^2/2 + A$, where $A \in \mathbb{R}$ is an arbitrary constant of integration. Hence we obtain the explicit solutions $\boxed{y(t) = \pm \mathrm{e}^A \mathrm{e}^{t^2/2}}$, all of which are defined for all t. Now we note that as A is a constant, then $B = \pm\mathrm{e}^A$ is also a constant, and the typical solutions can be written simply as $y(x) = B\,\mathrm{e}^{t^2/2}$. The difference between the two forms is that A is any real constant, $A \in \mathbb{R}$, but B, by construction, must be a nonzero constant, $B \in \mathbb{R} \setminus \{0\}$. The next observation is that the previously found special solution $y(t) \equiv 0$, can be produced by $y(x) = B\,\mathrm{e}^{t^2/2}$ with $B = 0$ (prohibited for the typical solutions). So in this example the two classes of solutions can be combined into a single formula,

$$\boxed{y(t) = B\,\mathrm{e}^{t^2/2}, \qquad B \in \mathbb{R}}\,,$$

in which $B = 0$ corresponds to the special solution and $B \neq 0$ corresponds to the typical solutions.

[iii]

For every point in the plane there is an integral curve passing through it, so any IVP is locally solvable. Also, there is no intersection between different integral curves, so there is no non-uniqueness. This agrees with the predictions of Peano's and Picard's theorems.

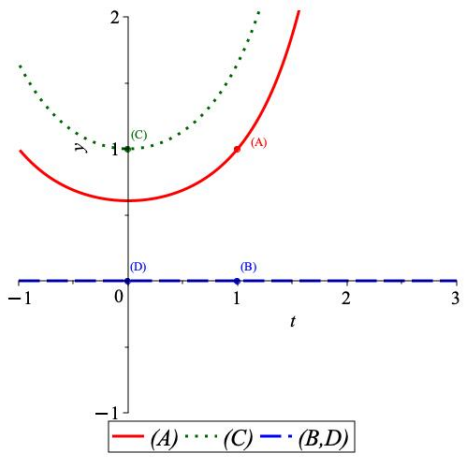


[iv] (A) We substitute $t = 1$, $y = 1$ into the general solution and get $1 = B\,\mathrm{e}^{1/2}$ hence $B = \mathrm{e}^{-1/2}$ and $\boxed{y(x) = \mathrm{e}^{(t^2-1)/2}}$. Similarly,

(B) $t = 1$, $y = 0$ gives $0 = B\,\mathrm{e}^{1/2}$, i.e. $B = 0$ and $\boxed{y(t) = 0}$;

(C) $t = 0$, $y = 1$ gives $1 = B\,\mathrm{e}^{0}$, i.e. $B = 1$ and $\boxed{y(t) = \mathrm{e}^{t^2/2}}$;

(D) $t = 0$, $y = 0$ gives $0 = B\,\mathrm{e}^{0}$, i.e. $B = 0$ and $\boxed{y(t) = 0}$, same as for (B). All of these solutions are defined and unique for all t.



(b) **Question** $y' = 2ty^2$, *(A)* $y(1) = 0$, *(B)* $y(0) = 1$.

Solution

[i] The right-hand side is $f(t, y) = 2ty^2$ and is continuous for all (t, y). The partial derivative $f_y(t, y) = 4ty$ is continuous for all (t, y). Therefore, the IVP will be locally solvable and locally uniquely solvable for any initial condition $\boxed{(t_0, y_0) \in \mathbb{R} \times \mathbb{R}}$.

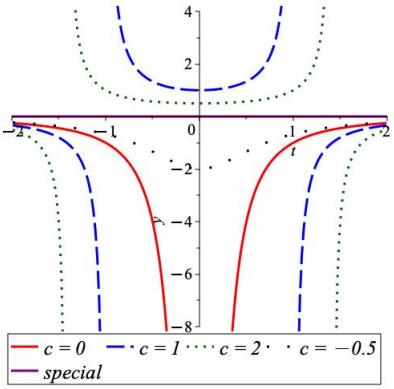
[ii] The right-hand sides factorises as $f(t, y) = 2ty^2 = F(t)G(y)$ with $F(t) = 2t$ and $G(y) = y^2$, for example. We have $G(y) = 0$ for $y = 0$, hence $\boxed{y(t) \equiv 0}$ is a special solution and it is defined for $\boxed{t \in \mathbb{R}}$. For $y \neq 0$, we get the typical solution from integrating $y^{-2}\,\mathrm{d}y = 2t\,\mathrm{d}t$, giving $-y^{-1} = -C + t^2$, where C is a constant (the minus sign being for later convenience). Then $\boxed{y(t) = \dfrac{1}{C - t^2}}$. This is only defined when $t^2 \neq C$, which means that

- for $C < 0$, it is defined $\boxed{t \in I = \mathbb{R}}$,
- for $C = 0$, this gives two solutions defined in two separate intervals, $\boxed{t \in I_1 = (-\infty, 0)}$ and $\boxed{t \in I_2(0, \infty)}$, and
- for $C > 0$, the same formula gives three solutions, defined correspondingly in the intervals $\boxed{t \in I_1(-\infty, -\sqrt{C})}$, $\boxed{t \in I_2 = (-\sqrt{C}, \sqrt{C})}$ and $\boxed{t \in I_3 = (\sqrt{C}, \infty)}$.

The formula for the typical solution cannot produce $y \equiv 0$ for any choice of the constant C, so the special solution and the typical solutions cannot be combined into a single formula.

[iii]

Here again, for every point in the plane there is an integral curve passing through it, so any IVP is locally solvable. Also, there is no intersection between different integral curves, so there is no non-uniqueness. This agrees with the predictions of Peano's and Picard's theorems.



[iv] (A) We substitute $t = 1$, $y = 0$ into the general solution and get $0 = 1/(C - 1)$ which does not have solutions. This initial condition is satisfied only by the special solution $\boxed{y = 0}$, which exists and is unique for all t, $\boxed{I = \mathbb{R}}$.

(B) We substitute $t = 0$, $y = 1$ into the general solution and get $1 = 1/(C-0)$, hence $C = 1$. The special solution does not fit here. The unique solution is therefore $\boxed{y = 1/(1 - t^2)}$. The denominator must be nonzero and the interval should include $t = 0$, so the interval where the solution is defined and unique is $\boxed{I = (-1, 1)}$.

